Advanced Biotechnology

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CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecture 4

Nucleic acid sequencing – Part II

Third generation sequencing and applications



UNIVERSITY OF CAMBRIDGE

Lecture outline

1) Nanopore sequencing
2) PacBio SMRT sequencing
3) Applications of nucleic acid sequencing
   - Market and economics
   - Sequencing applications

Learning outcomes

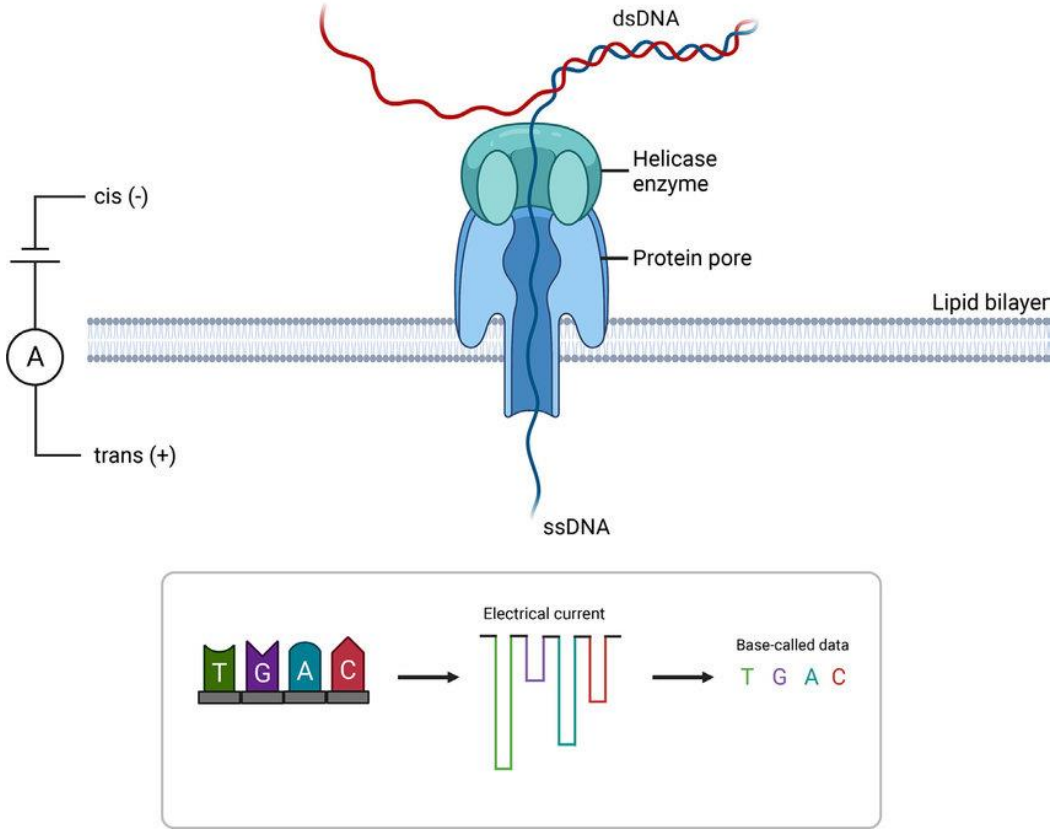
1) Describe the principles of Nanopore Sequencing
2) Describe applicability, limitations, advantages and disadvantages of Nanopore Sequencing
3) Describe principles of PacBio SMRT sequencing
4) Describe applicability, limitations, advantages and disadvantages of PacBio SMRT sequencing
5) Describe types of genome and transcriptome sequencing
6) Describe applications of sequencing to biotechnology, personalised medicine, disease screening and diagnosis, pre-natal screening, virus surveillance, environmental sequencing, forensics

Lecture outline

1) **Nanopore sequencing**
2) PacBio SMRT sequencing
3) Applications of nucleic acid sequencing
   - Market and economics
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Third generation sequencing: Nanopores

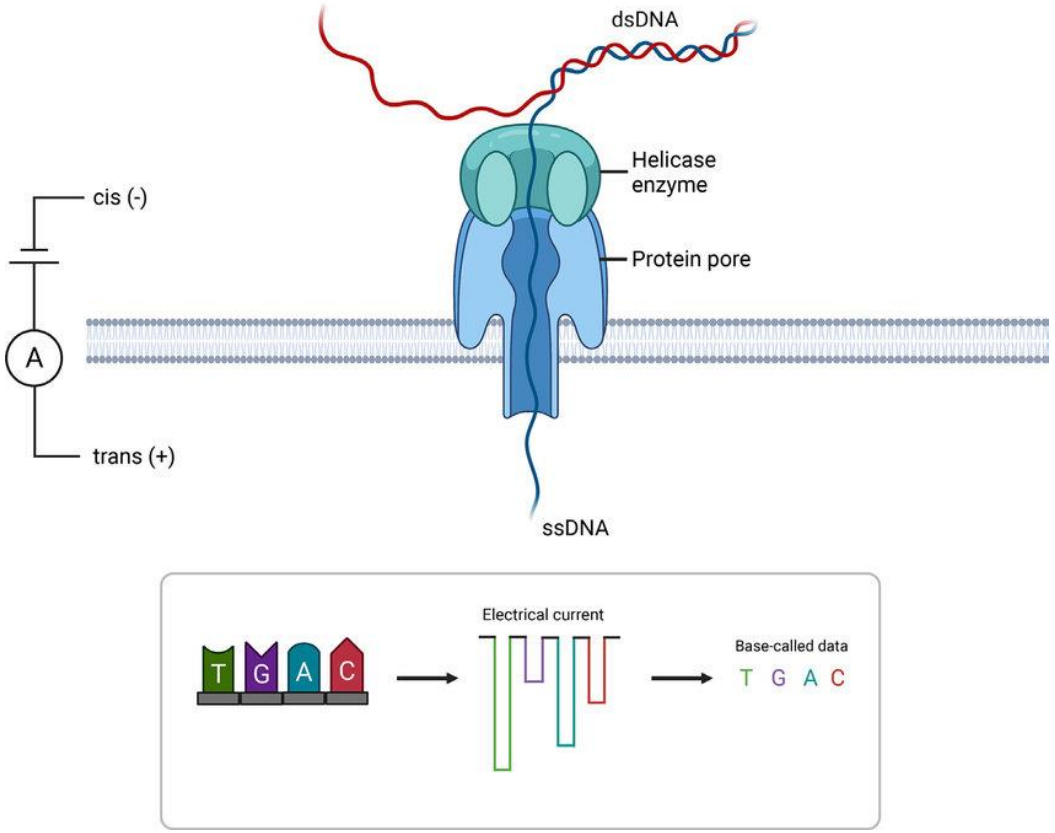
- **NGS requires amplification** because it is hard to detect fluorescence from individual fragments. This makes sample preparation cumbersome
- Additionally**, it is difficult with NGS to read long sequences**, which makes reconstruction of gene sequences more challenging and prone to error
- Third generation sequencing, and particularly Nanopore Sequencing addresses these issues



Third generation sequencing: Nanopores

Principle:

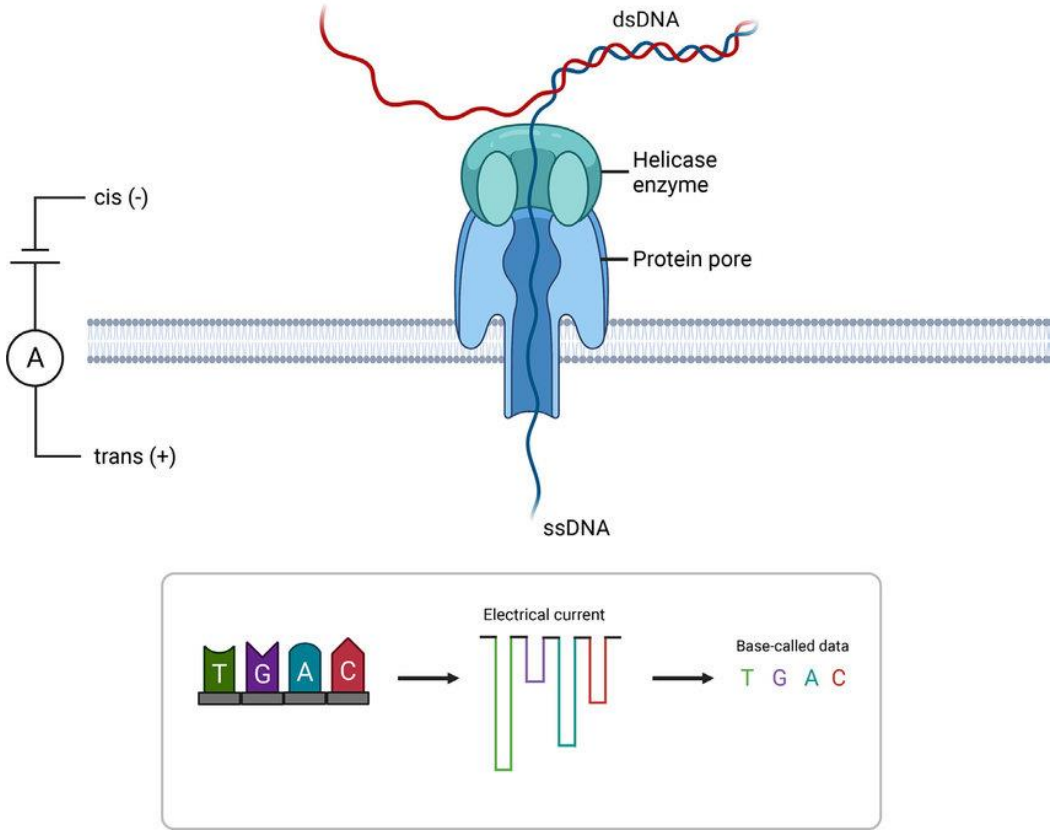
- Consider a small protein pore inserted in a membrane, with the width of the channel being similar to the width of single-stranded DNA
- Apply an electric potential across the membrane, which causes an ionic current (ions need to be present)
- The ionic current can be measured **very precisely**



Third generation sequencing: Nanopores

Principle:

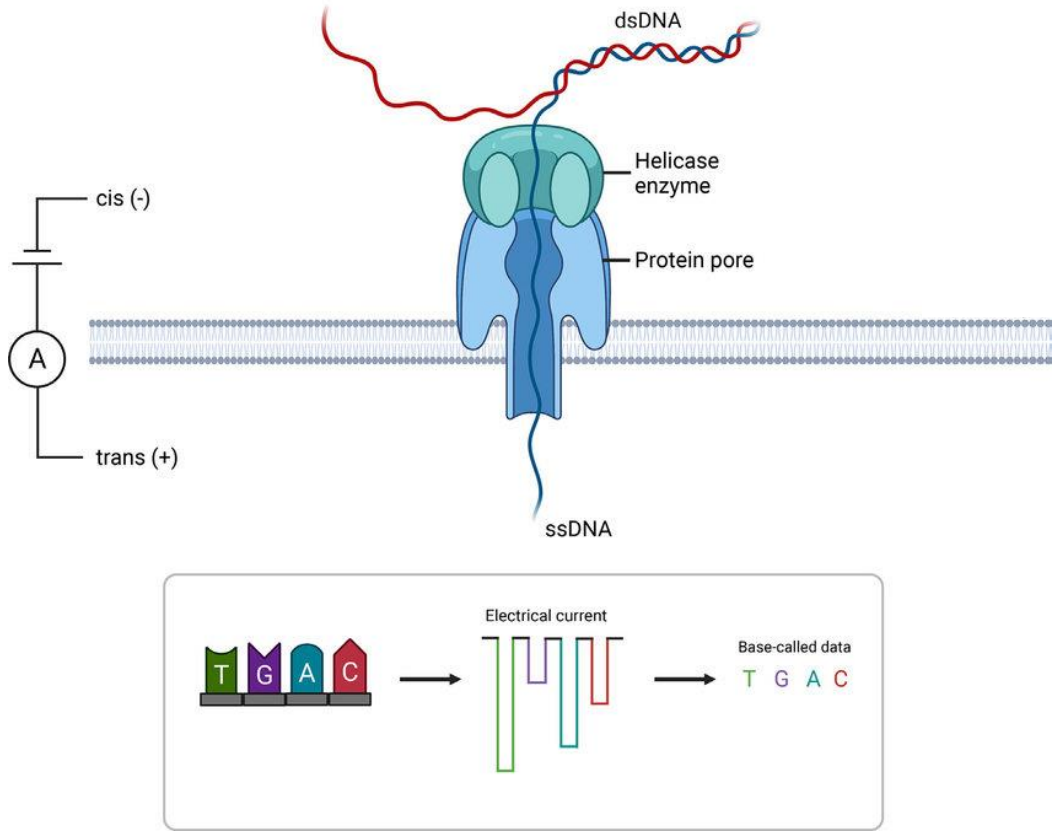
- Charged molecules (like DNA) will also be dragged through the pore
- Every time a molecule goes through the pore, it temporarily blocks some of the ions, and we can detect a dip in the ionic current
- The size, length and shape of the molecule can (with some effort) be inferred from the characteristics of this current dip (amplitude, duration etc)



Third generation sequencing: Nanopores

Principle:

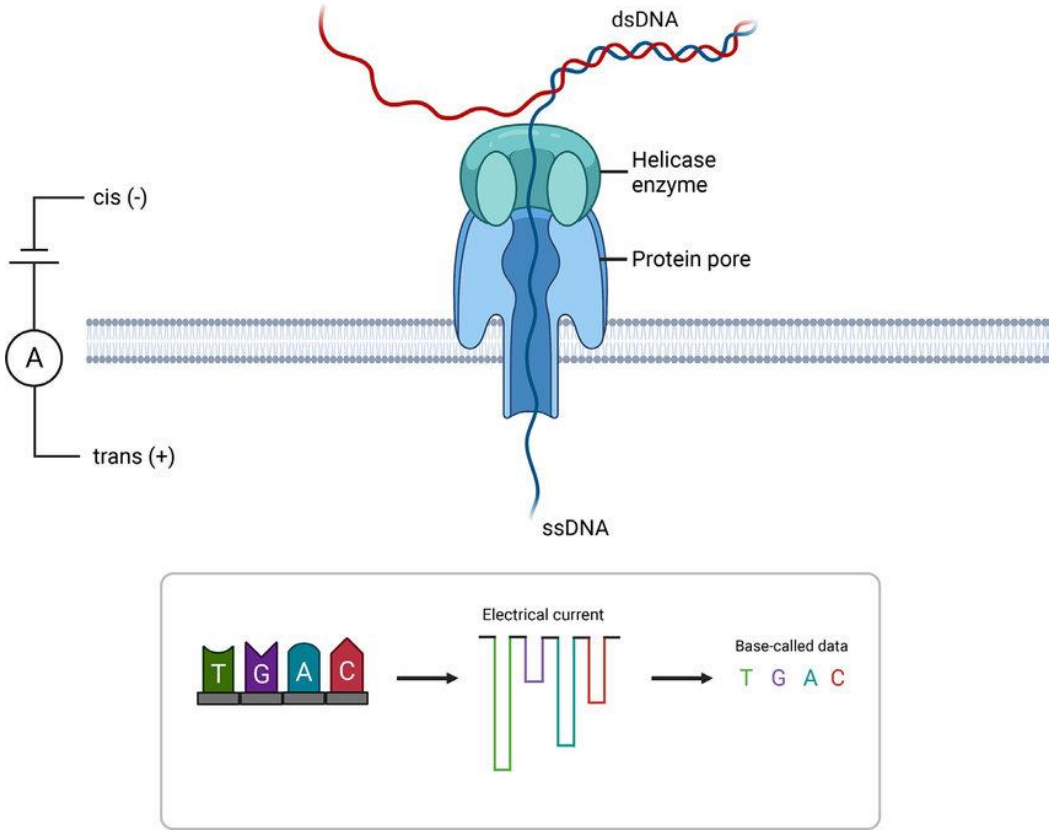
- When a single-stranded DNA molecule goes through the pore, each combination of nucleotides passing through the pore produces a slightly different dip in the ionic current, which we can use to identify the bases (base calling)
- This principle can thus be used for sequencing



Third generation sequencing: Nanopores

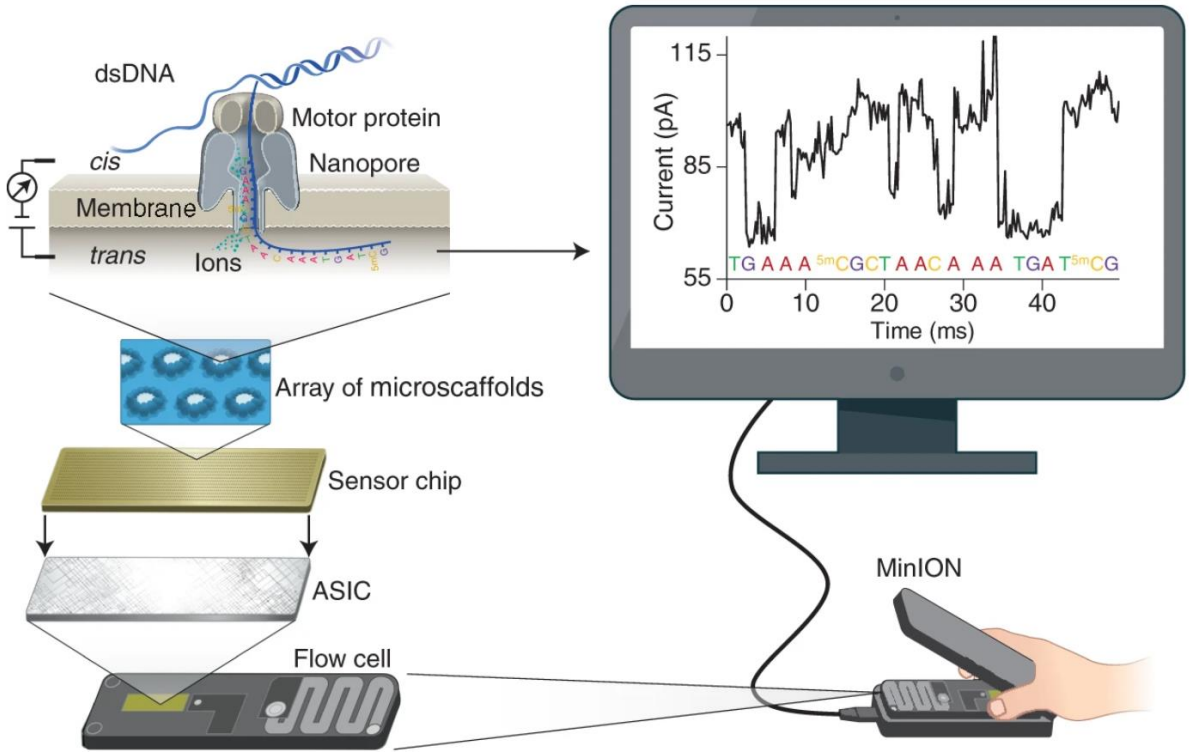
Principle:

- Protein pores used have a width of the channel similar to that of DNA, and are further engineered to enhance signal-to-noise ratio
- They are fused to a helicase enzyme, which serves two functions:
    1. It unwinds the double-stranded DNA, such that only one strand goes through the pore
    2. It slows down translocation to improve signal and facilitate base calling



Oxford nanopore technologies

- Spin-out from University of Oxford
- Portable handheld device for DNA/RNA sequencing
- Devices work by having an array of microwells, most of which contain a membrane with a single nanopore, enabling highly parallel sequencing



Credit: ONT

Nanopore sequencing: Pros and cons

Pros

- Ultra-long reads: can sequence fragments up to 100 kb
- Simple library prep and no amplification (no PCR). It takes minutes as opposed to days
- Can detect epigenetic markers
- Can sequence RNA directly
- Kit is portable and can be deployed for rapid diagnostics, even in the field (even in space)
- Kit is cheaper

Cons

- Low accuracy of the reads, in some cases down to 95% (vs 99.9% for NGS)
- Systematic errors. E.g. it is challenging to sequence homopolymers (e.g. CCCCC)
- Consumables (cartridges) are expensive, ~$1000 each

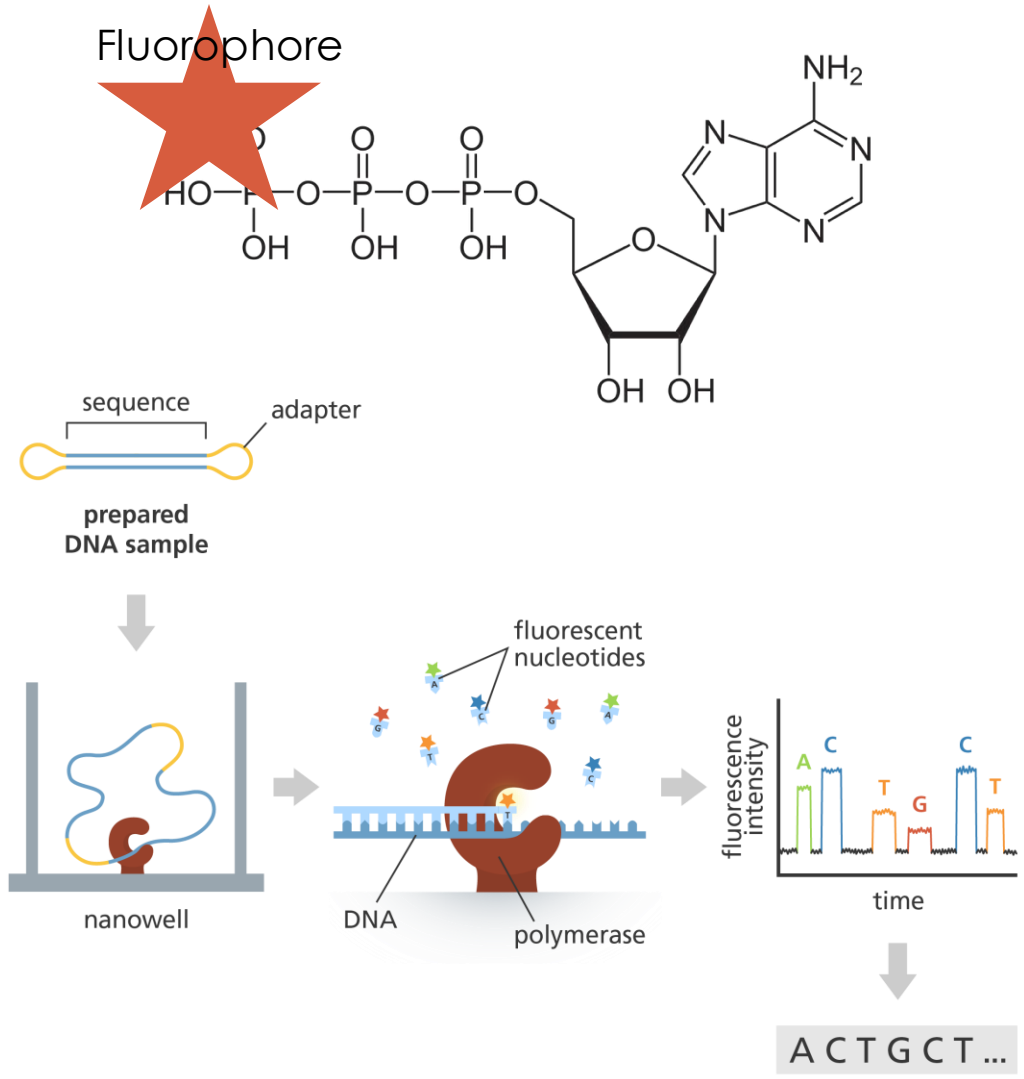
Lecture outline

1) Nanopore sequencing
2) **PacBio SMRT sequencing**
3) Applications of nucleic acid sequencing
   - Market and economics
   - Genomics
   - Sequencing applications

Third generation sequencing: PacBio Single Molecule Real Time (SMRT) sequencing

Principle:

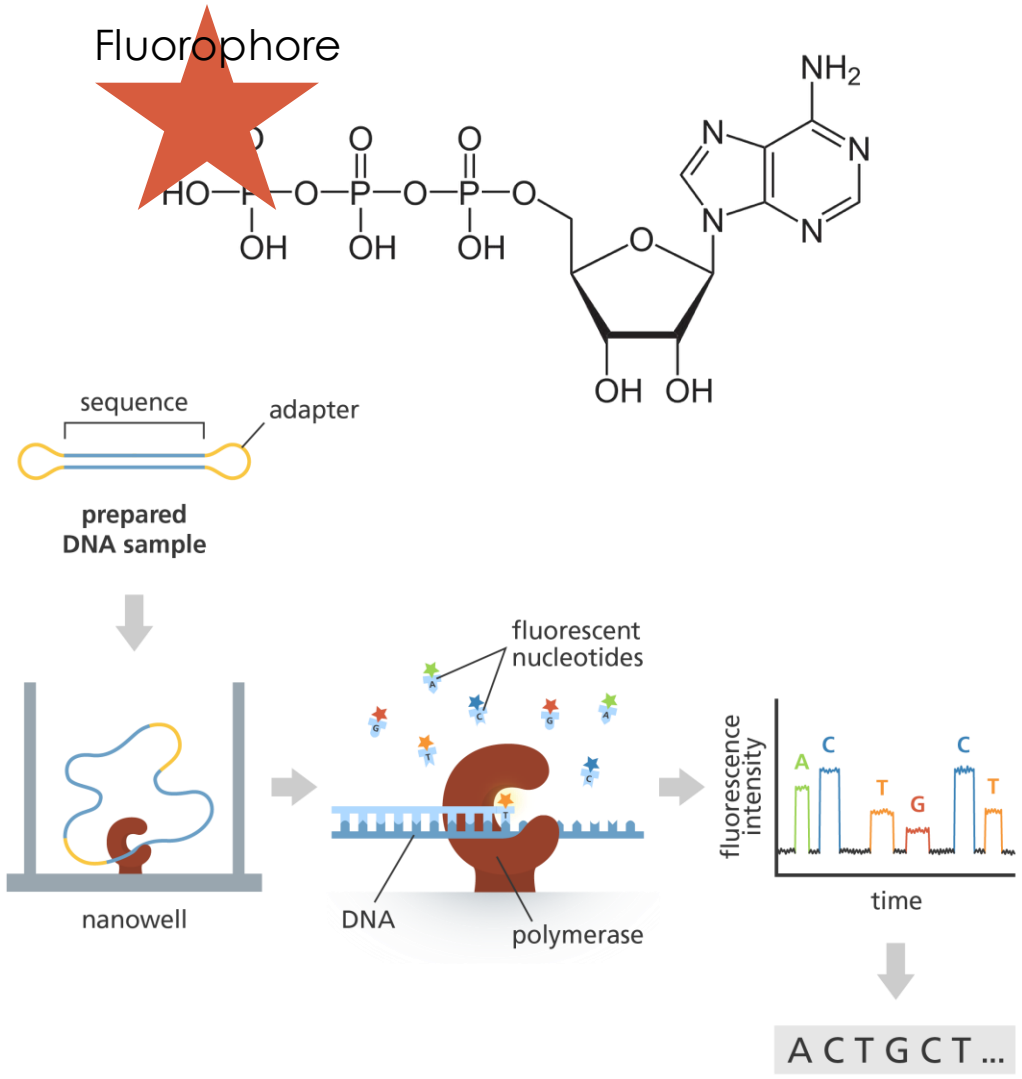
- Library preparation involves adding **hairpin adapters to circularise** the fragments
- The instrument uses a large array of nano wells, each with a single DNA polymerase attached to it, which captures a single circular fragment
- Fluorescent dNTPs are included, which are **fluorescently labelled on the phosphate group**



Third generation sequencing: PacBio Single Molecule Real Time (SMRT) sequencing

Principle:

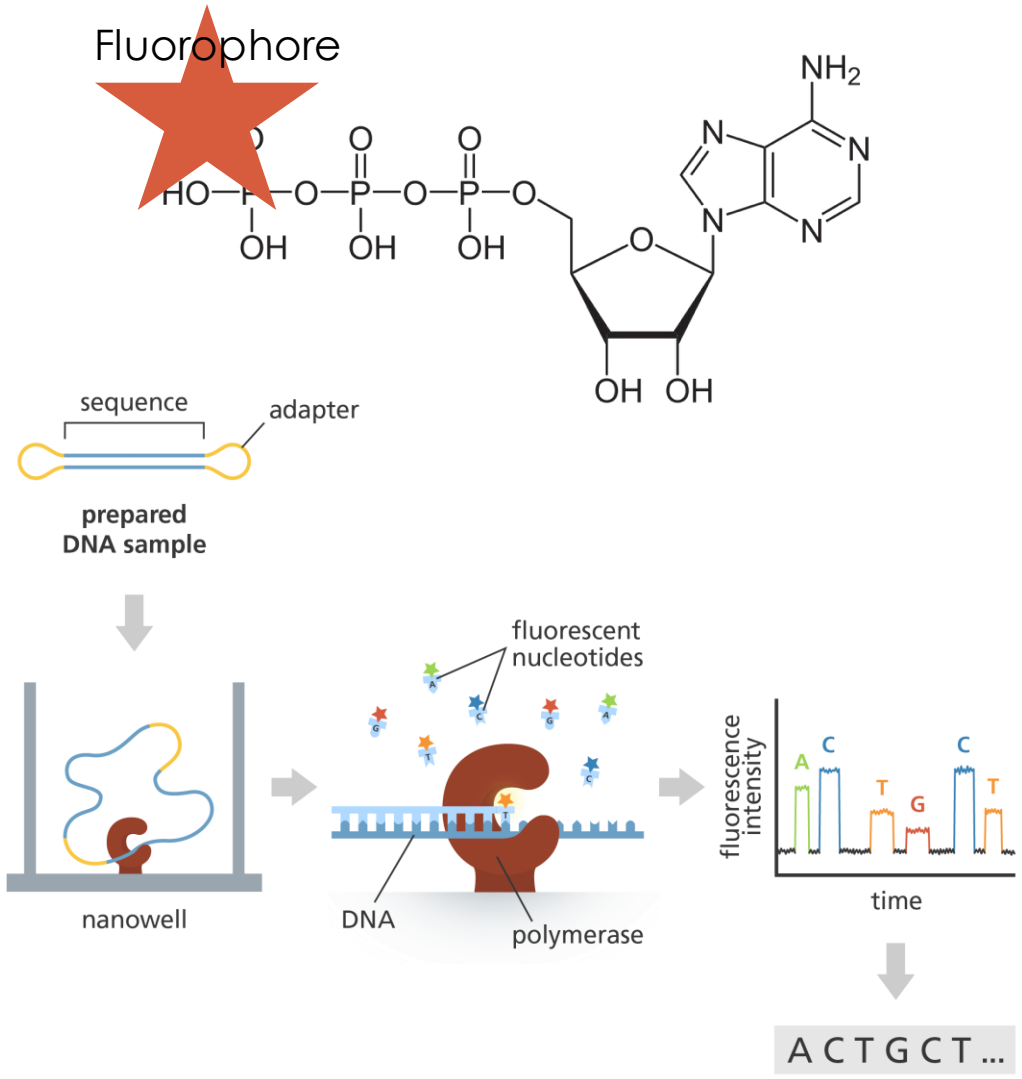
- The polymerase starts copying the circular DNA
- Every time the polymerase adds one dNTP, the fluorescent molecule is **transiently immobilised** within then nano well, and can be detected (**before being released**)
- Detection uses **Zero Mode Waveguides**, which enables the signal to be collected only from the bottom of the nano well, minimising noise from other dNTPs



Third generation sequencing: PacBio Single Molecule Real Time (SMRT) sequencing

Principle:

- DNA sequence can then be reconstructed from the sequence of fluorescence signal detected in a given nano well
- A chip of used for SMRT contains millions of nano wells, enabling high throughput



Third generation sequencing: PacBio Single Molecule Real Time (SMRT) sequencing

- Technology developed by Pacific Bioscience (PacBio) in 2009



Credit: PacBio

PacBio sequencing: Pros and cons

Pros

- Long reads, 10-25 kb (but not quite as long as nanopore)
- Highly accurate, up to 99.9%
- No amplification required
- Does well with repetitive sequences and homopolymers (contrary to nanopore)

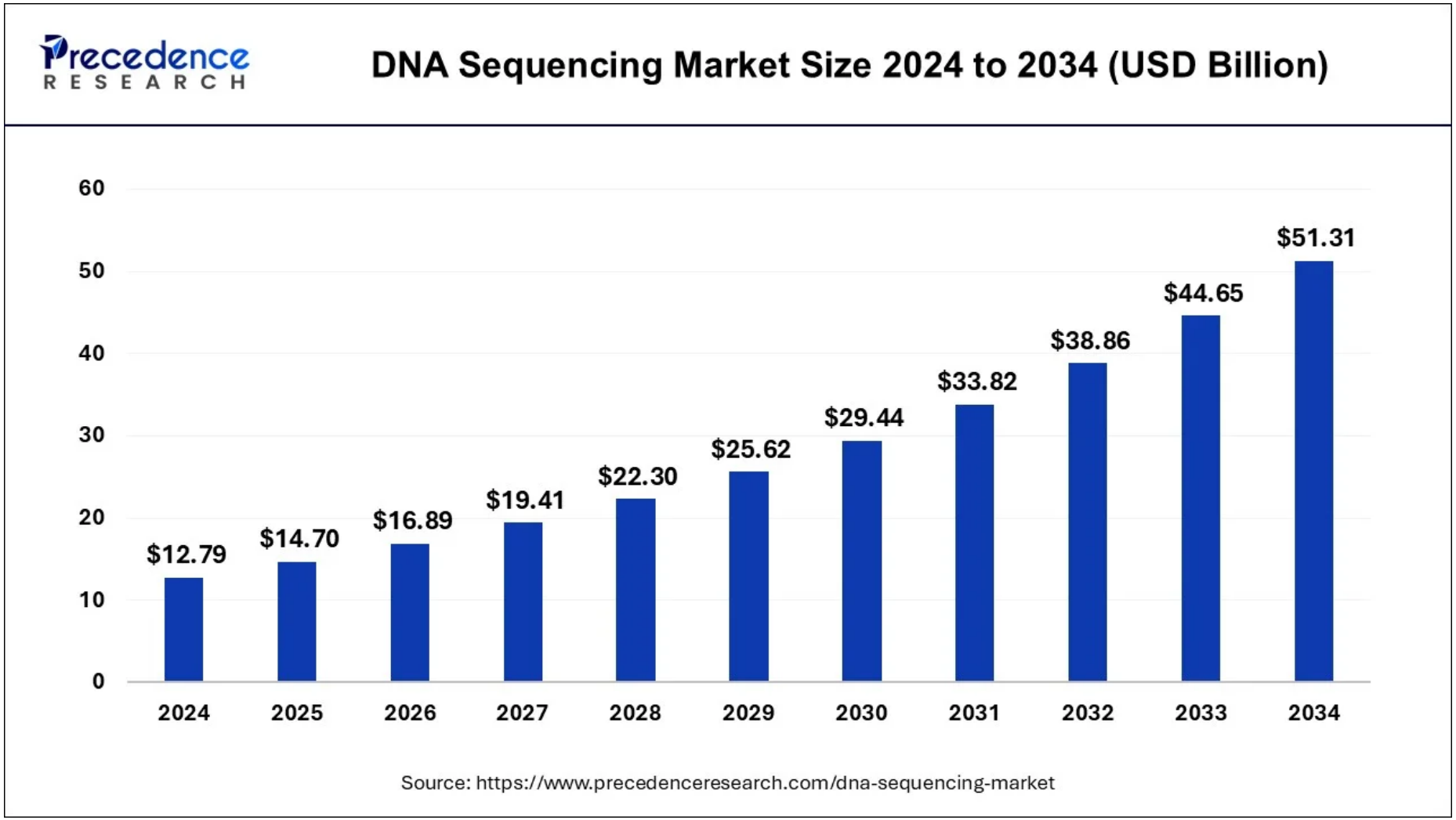
Cons

- Library preparation (circularisation) required
- High cost per nucleotide compared to nanopore and Illumina
- Instrument expensive and large (not portable)
- Long experiment time (15-30 hours)

Lecture outline

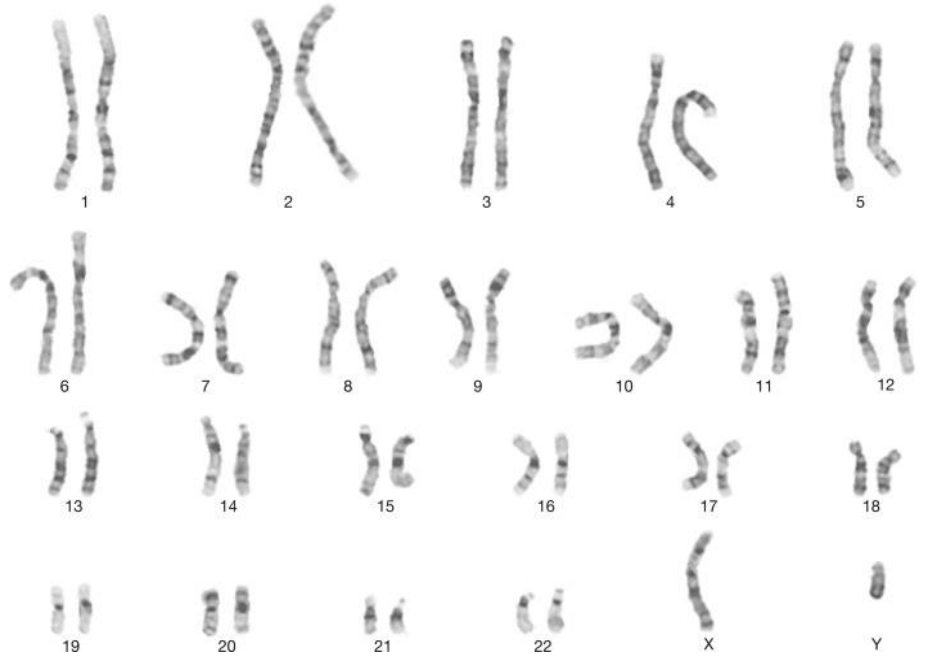
1) Nanopore sequencing
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   - Sequencing applications

Market value



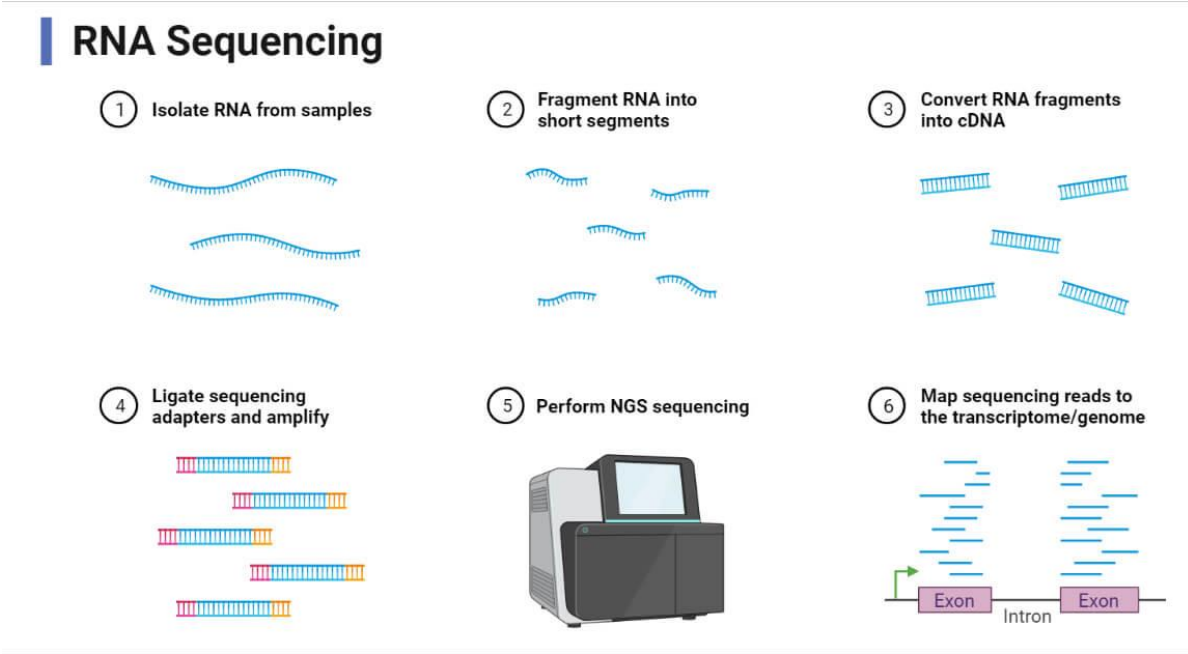
Types of genome sequencing

1) **Whole genome sequencing**
   - Sequence the entire genome
   - Most informative, but expensive
   - Useful for detecting rare variants, structural changes, and non-coding mutations
2) **Whole exome sequencing**
   - Sequence only the protein-coding regions (exome)
   - 1-2% of human genome
   - More cost-effective
   - Most disease-related gene variants are in exomes
3) **Targeted genome sequencing**
   - Sequence specific regions of the genome (e.g. cancer-related genes; inherited disease genes)
   - Customizable based on clinical or research needs
   - Requires prior knowledge to select target regions



Transcriptomics (RNA-sequencing)

- Sequence RNA molecules
- RNA is normally not sequence directly. It is reverse-transcribed into DNA with reverse transcriptase
- Reveals
  - Gene expression levels
  - Alternative splicing patterns
- Example Applications
  - Study cellular responses to drugs or environmental stimuli
  - Profiling gene expression in cancer cells to identify biomarkers or therapeutic targets
  - Investigating developmental biology, immune responses, or disease progression

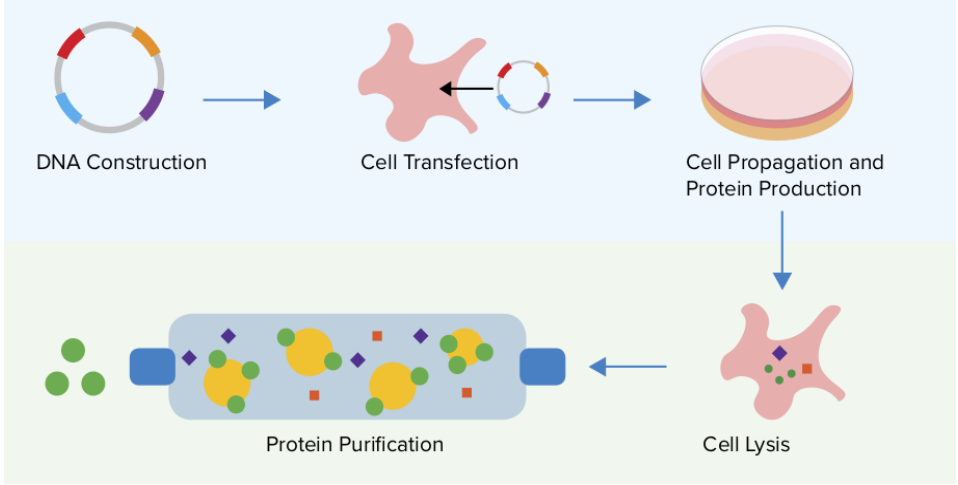


Examples of Sequencing Applications

1) Biotechnology workflows
2) Personalized medicine
3) Detection of genetic diseases
4) Risk prediction
5) Pre-natal screening
6) Genetic surveillance of viruses
7) Environmental sequencing
8) Forensics

Engineering new genetic sequences

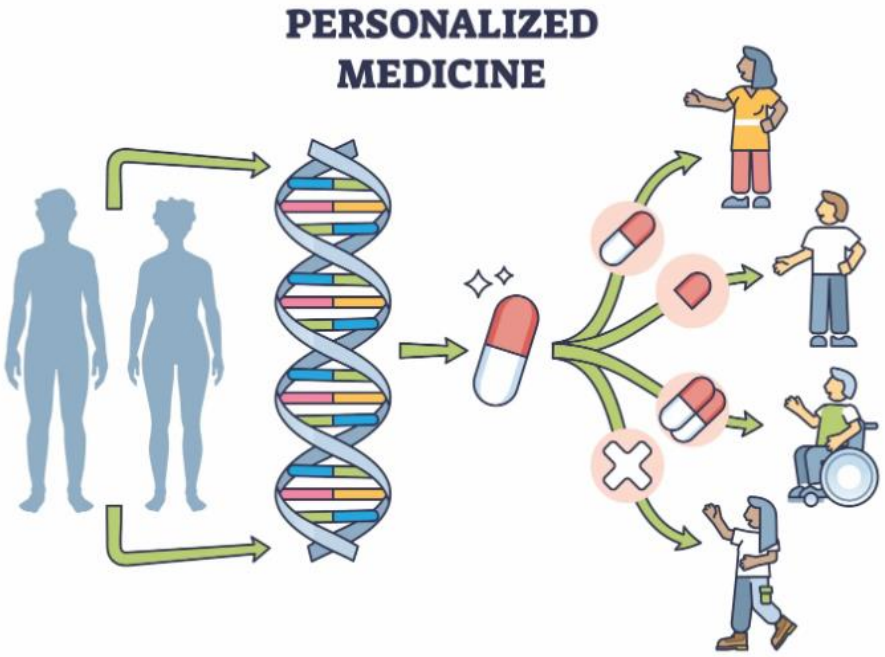
- Many biotechnology advances rely on engineering new genetic sequences (e.g. plasmid, bacteria, plants, animals) - You have learned this!
- Sequencing is essential to verify the intended modifications
- Confirms
  - Correct insertion of genes
  - That gene are not mutated
  - Absence of off-target effects





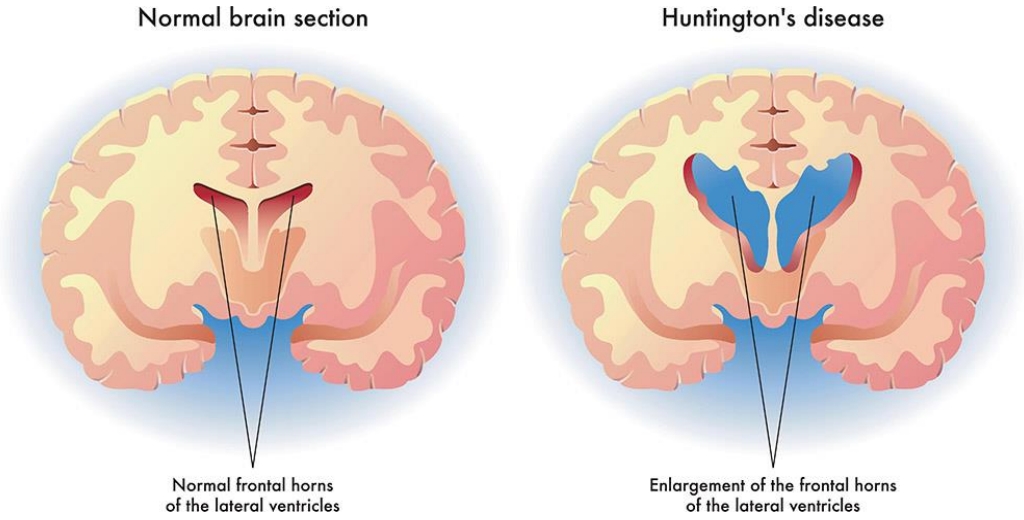
Personalized medicine

- Personalized medicine: same disease, same symptoms, different treatments.
- Use gene profiles to guide therapy
- Example treatment of non-small cell lung cancer
  - Test tumour tissue for mutations
  - If EGFR mutation is detected, treat with EGFR inhibitor
  - If ALK mutation is detected, treat with ALK inhibitor.



Detection of rare genetic diseases

- Whole genome/exome sequencing for undiagnosed cases. Many genetic diseases are otherwise hard to diagnose
- Examples:
  - Huntington's disease (CAG repeat expansion)
  - Duchenne muscular dystrophy (DMD gene mutations)
- Applications
  - Diagnosis for targeted therapies
  - Carrier screening for inherited conditions
  - Family planning and reproductive decision-making



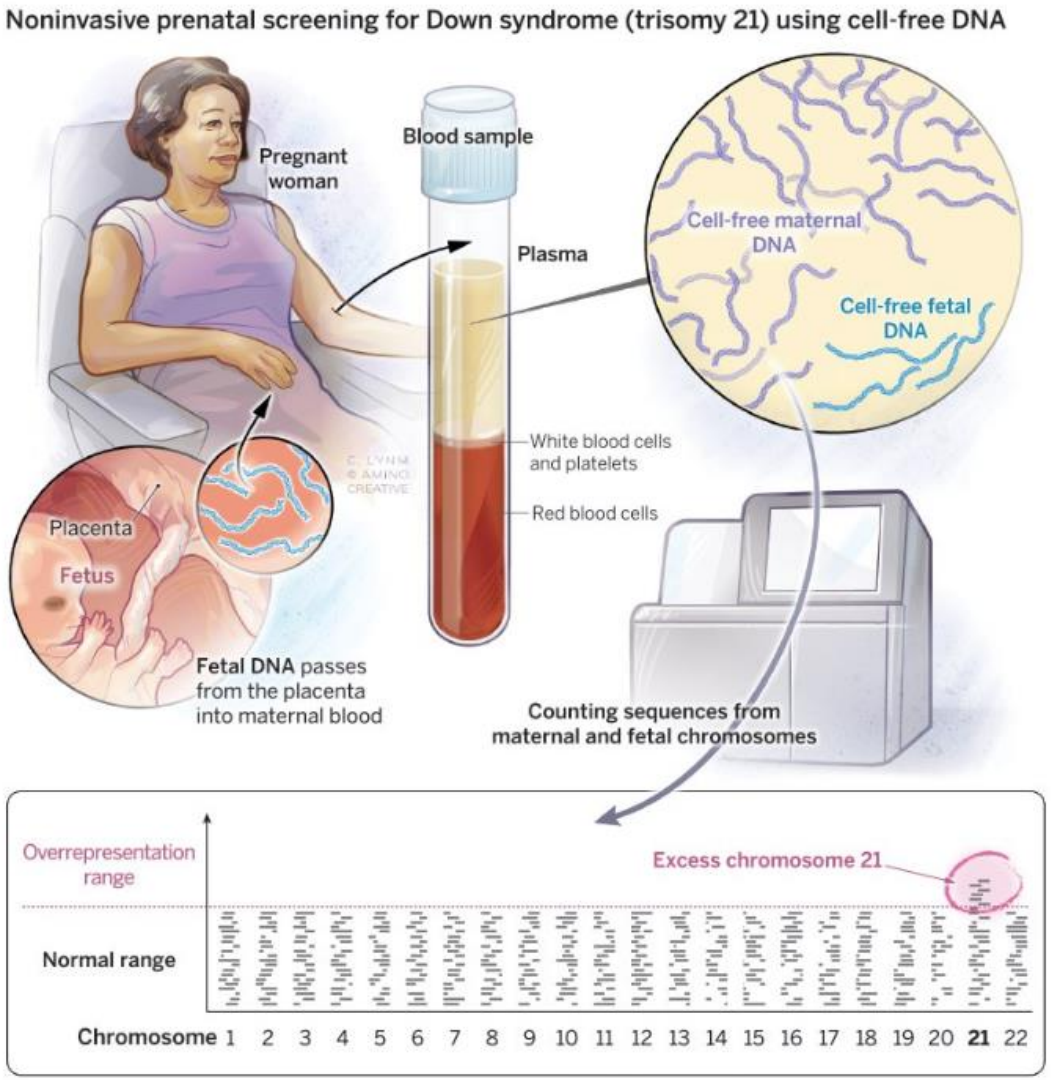
Genetic disease risk prediction

- Diseases like type 2 diabetes and coronary artery disease are influenced by thousands of genetic variants
- Sequencing identifies high-risk individuals before symptoms appear
- Enables
  - Earlier screening
  - Lifestyle changes
  - Preventive therapies



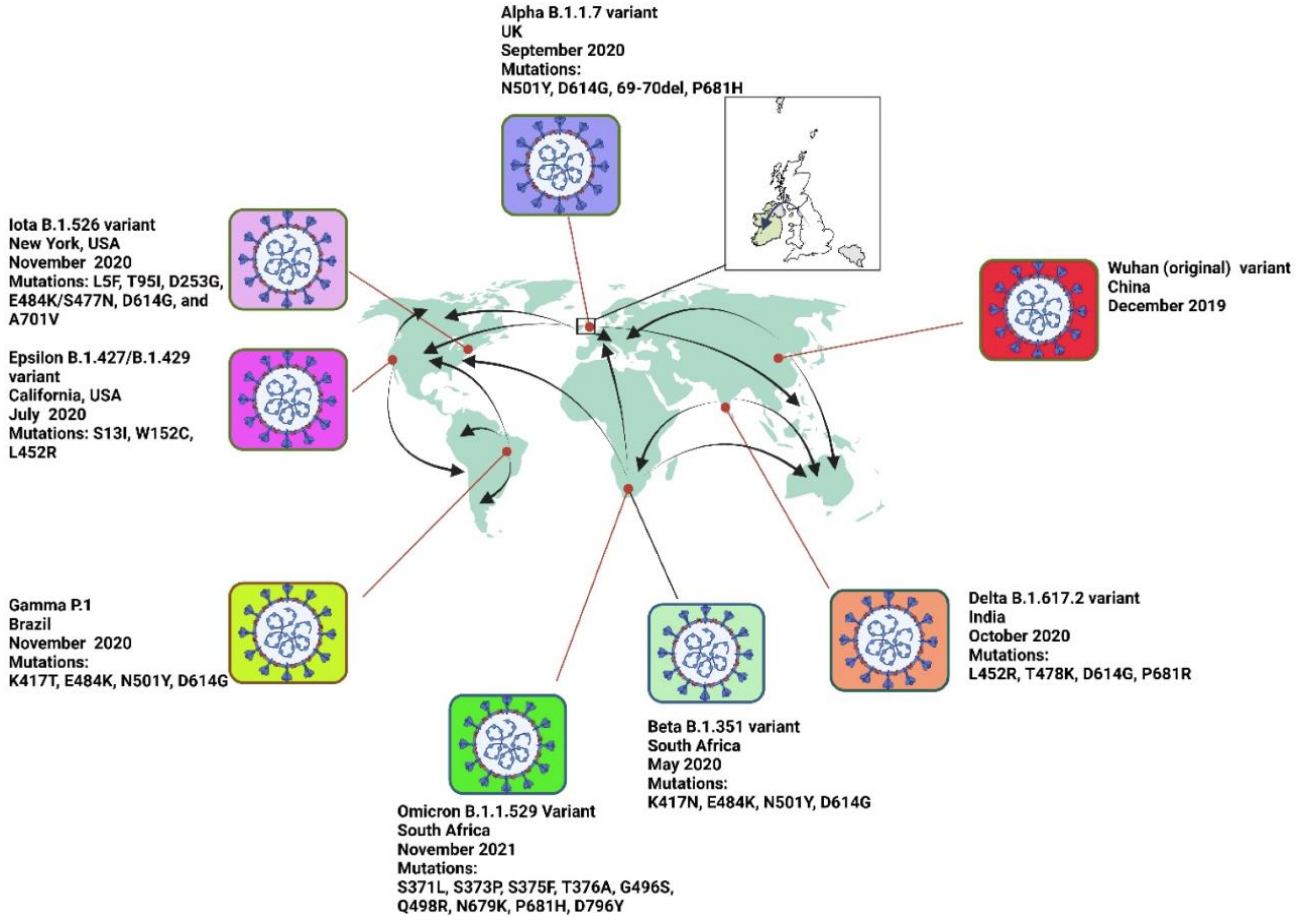
Pre-natal screening of foetal DNA

- Foetal DNA can be found in maternal blood
- Sequencing DNA from maternal blood detects genetic disorders in the foetus
- Example: Down syndrome: extra copy of chromosome 21 (how does sequencing identifies extra copies of chromosomes?)
- Pre-natal screening can be used to detect several conditions (e.g. cystic fibrosis, sickle cell disease…)



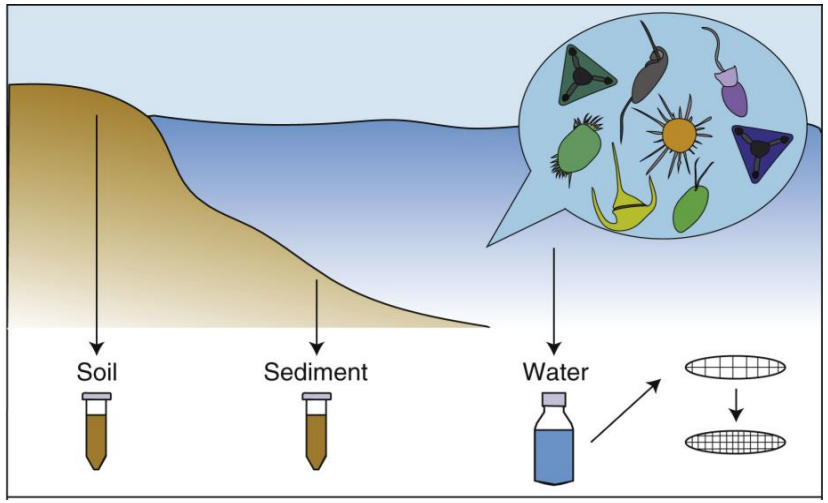
Genomic surveillance of viruses

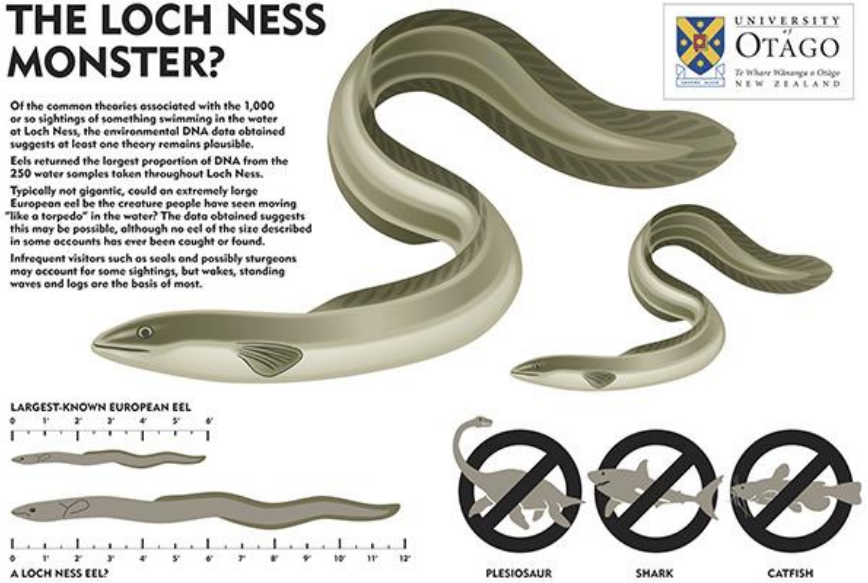
- Sequence viral genomes to monitor mutations
- Applications
  - Identify variants of concern
  - Track transmission chains
  - Support vaccine updates (covid, flu…)



Environmental DNA (eDNA) sequencing

- Sequence all DNA in water/soil samples
- You can find DNA from everything that lives in an environment: microorganisms, animals, plants, fungi…
- Applications
  - Learn about makeup of microbial communities
  - Figure out if a given specie lives somewhere
  - Discover new species or sub-groups
    - Fish:https://www.nhm.ac.uk/discover/news/2024/february/new-technique-detects-distinct-fish-populations-in-a-single-lake.html
    - Daisies:https://www.gu.se/en/news/new-species-discovered-with-refined-dna-technology
    - Beetles:https://royalsocietypublishing.org/doi/10.1098/rspb.2015.2350
    - Loch Ness monster: https://www.bbc.co.uk/news/uk-scotland-highlands-islands-49495145





Forensics

- Collect DNA from hair, blood, saliva, semen, remains…
- Identifying criminals by matching with crime-scene DNA
- Exonerate suspects
- Kinship and paternity testing
- Identification of remains

